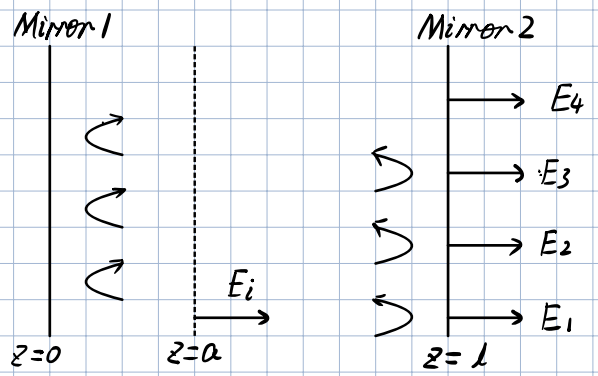


$$E_{out} = E_1 + E_2 + E_3 + \cdots$$

$$E_1 = t_2 e^{-ik'(l-a)} E_i$$

$$E_2 = (r_1 r_2 e^{-i2k'l}) t_2 e^{-ik'(l-a)} E_i$$

$$E_3 = (r_1 r_2 e^{-i2k'l})^2 t_2 e^{-ik'(l-a)} E_i$$

$$E_4 = (r_1 r_2 e^{-i2k'l})^3 t_2 e^{-ik'(l-a)} E_i$$

$$E_{out} = t_2 e^{-ik'(l-a)} E_i \left[1 + (r_1 r_2 e^{-i2k'l}) + (r_1 r_2 e^{-i2k'l})^2 + \cdots\right]$$

$$= \frac{t_2 e^{-ik'(l-a)}}{1 - r_1 r_2 e^{-i2k'l}} E_i$$

k': 复传播常数. $\quad k' = k_0 + \Delta k + i(g-a)/2$.

$$E_{out} = \frac{t_2 e^{-i[(k_0+\Delta k)(l-a)]} e^{(g-a)(l-a)/2}}{1 - r_1 r_2 e^{-i2(k_0+\Delta k)l} e^{(g-a)}} E_i$$

$$r_1 r_2 e^{-i2(k_0+\Delta k)l} e^{(g-a)} = 1$$